

The project is a Marie Skłodowska-Curie Doctoral Network that trains a new generation of researchers to develop trustworthy, causal and interoperable AI for cardiovascular healthcare. Current medical AI systems often lack generalisability, transparency and clinical validation. The project addresses these challenges by integrating leading European universities, hospitals and industry partners into a coherent programme that connects methodological innovation with real-world clinical relevance and regulatory readiness. The network is structured around four scientific areas: causal and explainable modelling, multimodal data integration, human-in-the-loop learning, and clinical evaluation through federated and retrospective trials. Doctoral Candidates will develop new computational methods, open-source tools and interoperability standards aligned with the EU AI Act, FAIR data principles and the European Health Data Space. Their research is embedded in cross-sector environments, supported by tailored supervision and guided by Individual Training Panels and Career Development Plans. A three-tier training programme combines local PhD curricula with network-wide schools, mobility-rich secondments and creation of the European Causal AI Academy, ensuring long-term access to learning materials, datasets and software. Training in open science, ethics, data governance and innovation equips graduates for careers across academia, clinical research, industry and regulation. By producing scientific advances, robust evaluation frameworks and highly skilled researchers, The project will accelerate the uptake of reliable, human-centred AI technologies in cardiovascular medicine. Its outcomes aim to strengthen patient safety, support evidence-based decision making and contribute to Europe's leadership in secure, transparent and sustainable healthcare AI.

#### Evaluation Summary Report

Evaluation Result Total score: 85.80 % (Threshold: 70 /100.00)

Criterion 1 - Excellence Score: 4.20 (Threshold: 3 / 5.00 , Weight: 50.00% )

- Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art).
- Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality and appropriateness of open science practices).
- Quality and credibility of the training programme (including transferable skills, inter/multidisciplinary, inter-sectoral and gender as well as other diversity aspects).
- Quality of the supervision (including mandatory joint supervision for industrial and joint doctorate projects).

The proposal presents clear objectives to develop AI-technologies for cardiovascular healthcare, with concrete measures that make them overall verifiable and achievable within the project's timeframe. The individual projects of the recruited researchers are appropriately articulated and convincingly integrated to support and strengthen the coherent outcomes of the research programme. The project goes beyond the state of

the art in cardiovascular care and it is ambitious. Overall, the methodology is clearly described, including the underlying concepts and models are mostly detailed. Several aspects of interdisciplinary approaches are coherent, and well-integrated to carry out the objectives of the proposal. However, AI/ML plans- specifically conceptual paradigms, methods, and validation metrics, as well as critical security aspects for handling clinical data with AI are insufficiently detailed. It is also not fully clear how some of the chosen methods will deliver key objectives: cardiovascular AI models are evaluated only retrospectively with limited plans for prospective validation; the cohort is not sufficient concerning the multihospital, multi-operator design needed to reflect contemporary European TAVI practice to capture the diversity; and the number of contributing clinicians to the database is not well-specified. Together, these constitute a shortcoming. Gender and other diversity characteristics relevant to this research are mostly appropriately considered in the research methodology as scientific variables (disaggregated data, subgroup calibration, bias mitigation), however, they are not detailed enough regarding their essential role in clinical issues in cardiology. This is a minor shortcoming. Open science practices, such as preprints, and repositories, are included and adequately described in the proposed methodology. In general, a solid training framework with meaningful network-wide events - supporting well-structured scientific and transferable training - is described in the proposal. The training programme effectively integrates intersectoral and transferable skills through training schools, courses, conferences and secondments. While network-wide training overall complements the local training programmes, there is a slight limitation in clarity and depth in systematic description of complementation of local training. In addition, some details for training are not properly described, for example, the integration of entrepreneurship training lacks clear mechanisms for practical application. These constitute a minor shortcoming. The supervisors are experienced and highly qualified in their scientific roles. They demonstrate appropriate international and local experience across all relevant areas and facilitating the adequate and effective completion the individual research projects. The well-structured and multi-layered approach of supervision practices is solid and overall sound. It ensures effective progress monitoring, feedback and support for the fellows; for example, well-defined and reviewed Personal Career Development Plans ensure systematic tracking of training objectives. However, the description remains slightly generic on how the quality of supervisory practices will be measured. This is minor shortcoming.

Criterion 2 - Impact Score: 4.50 (Threshold: 3 / 5.00 , Weight: 30.00%)

- Contribution to structuring doctoral training at the European level and to strengthening European innovation capacity, including the potential for: a) meaningful contribution of the non-academic sector to the doctoral training, as appropriate to the implementation mode and research field b) developing sustainable elements of doctoral programmes.
- Credibility of the measures to enhance the career perspectives and employability of

researchers and contribution to their skills development. • Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities. • The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts.

Non-academic sector contribution to doctoral training is well described and compelling; they have a meaningful and tangible role through structured secondments in hospitals, SMEs, and regulatory bodies, ensuring hands-on exposure to clinical workflows, compliance, and innovation. Industry and regulatory partners actively co-supervise doctoral candidates (DCs), participate the Individual Training Panels, and the Supervisory Board, and lead network-wide training sessions. Several elements are convincingly sustained after the project; these include e.g. the development of a virtual knowledge hub. The robust strategies and mechanisms described ensure a sustained long-term collaboration, and foster future international and intersectoral research and cooperation. The project provides a robust and well-structured framework and environment to equip fellows with both technical and transferable skills, significantly enhancing and ensuring the future employability in both academia and industry. Specifically, the profile of fellows in trustworthy AI for clinical issues is highly relevant for hospitals, industry R&D departments, start-ups and research infrastructures. Transferable skills are integrated in the training, and intersectoral secondments and co-supervision expose fellows to hospitals, SMEs, and regulatory bodies, ensuring practical experience and networking. The measures to enhance the career perspectives of fellows are highly relevant and comprehensive and designed to prepare fellows for multidisciplinary careers. The dissemination, exploitation, and communication measures are comprehensive, and well-structured, with clear objectives and monitoring through measurable indicators. The target groups for these measures are clearly identified. The outline of intellectual property management strategy is mostly appropriately addressed. However, suitable strategies to protect industrial and other relevant outcomes are not convincingly elaborated; this is a minor shortcoming. The expected scientific impact is highly credible with clear breakthroughs in causal modelling. For example, the integration of regulatory-grade pipelines aligned with regulatory frameworks adds robustness. Indicators, such as cross-site transportability, make the scientific outcomes measurable and realistic. Societal impact is well-justified: effective and safe integration of AI in healthcare and regulatory frameworks will largely affect the quality of the cardiovascular care provided to patients, and is addressing patient benefit, healthcare-system resilience, and policy uptake. Economic and technological impacts, such as validated causal AI and interoperability with European Health Data Space/Fast Healthcare Interoperability Resources (EHDS/FHIR), and human capital development through multi-sector training, are overall valid, especially in

a long-term, however, the magnitude in terms of specific measures is insufficiently presented in the proposal. This is a minor shortcoming.

Criterion 3 - implementation Score: 4.20 (Threshold: 3 / 5.00 , Weight: 20.00% ) • Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages. • Quality, capacity and role of each participant, including hosting arrangements and extent to which the consortium as a whole brings together the necessary expertise. The work plan is overall of good quality. The work packages (WPs) are clearly aligned with the project objectives, and the allocation of WPs and tasks is coherent. Each WP includes generally well-defined and appropriate deliverables and milestones. However, several aspects lack appropriate detail such as the description of dependencies between work packages. In addition, the validation strategy for certain deliverables is not fully specified and the definitions and timing of some milestones is not well aligned with the overall proposal. These issues together are a shortcoming. Risk management is present and generally credible, with several appropriate mitigation measures. However, certain descriptions of risks remain generic and oversimplified, without measurable thresholds for activation of the mitigation. This is a minor shortcoming. The participating organisations have the required high-quality infrastructure and capacity to implement research of high quality and their tasks. The international consortium is well-structured, integrating academic and non-academic partners with expertise, complementary strengths, and interdisciplinary knowledge, matching with the project's objectives, and reinforcing the structural and functional capability of the Doctoral Network. Hosting arrangements via accredited doctoral schools and international offices are credible and supportive. The roles of all associated partners are described in detail